Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: King Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
5524 S Central AVE	Turlock	Stanislaus	95380

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 11/01/1962

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X057-X010-X009-XXXX

B. OPERATORS

King, David

Operator name: King, David

Telephone no.: (209) 761-5676 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
5524 S Central AVE	Turlock	CA	95380

This operator is responsible for paying permit fees.

C. OWNERS

King, David

Legal owner name: King, David

Telephone no.: (209) 761-5676 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
5524 S Central AVE	Turlock	CA	95380

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	32	44	70	22	8
Number under roof	276	0	0	0	0	0
Maximum number	276	32	44	70	22	8
Average number	276	32	44	70	22	8
Avg live weight (lbs)	1,400	1,450	800	600		

Predominant milk cow breed: Holstein

Average milk production: 70 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 8,722.19 *tons per reporting period*

Total nitrogen from manure: 110,756.51 *lbs per reporting period*

After ammonia losses (30% loss applied): 77,529.56 *lbs per reporting period*

Total phosphorus from manure: 18,380.81 *lbs per reporting period*

Total potassium from manure: 52,856.55 *lbs per reporting period*

Total salt from manure: 137,313.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 4,678,385 *gallons*

Total nitrogen generated: 15,923.75 *lbs*

Total phosphorus generated: 2,198.12 *lbs*

Total potassium generated: 29,289.39 *lbs*

Total salt generated: 189,128.64 *lbs*

	4,678,385	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	4,678,385	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
T.I.D Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
#1	20	20	2	both	X057-X010-X009-XXXX
#2	16	16	2	both	X057-X010-X010-XXXX
#3	11	11	2	both	X057-X010-X009-XXXX
#4	20	20	2	manure	X057-X025-X014-XXXX
Totals for areas that were used for application	67	67	8		
Totals for areas that were not used for application					
Land application area totals	67	67	8		

B. CROPS AND HARVESTS

#1

Field name: #1

10/18/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 10/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	251.48 *ton*	As-is		64.0	5,500.00	1,200.00	8,800.00		11.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.57	138.31	30.18	221.30	1,023.02

05/31/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 05/31/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/22/2023	479.14 *ton*	As-is		66.2	4,300.00	600.00	5,400.00		7.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.96	206.03	28.75	258.74	1,149.84

#2

Field name: #2

10/18/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 16 Plant date: 10/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	201.19 *ton*	As-is		64.0	5,500.00	1,200.00	8,800.00		11.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.57	138.32	30.18	221.31	1,023.05

05/31/2023: Corn, silage

Crop: Corn, silage Acres planted: 16 Plant date: 05/31/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/22/2023	383.31 *ton*	As-is		66.2	4,300.00	600.00	5,400.00		7.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.96	206.03	28.75	258.73	1,149.83

#3

Field name: #3

10/18/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 11 Plant date: 10/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	138.31 *ton*	As-is		64.0	5,500.00	1,200.00	8,800.00		11.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.57	138.31	30.18	221.30	1,022.99

#3

05/31/2023: Corn, silage

Crop: Corn, silage Acres planted: 11 Plant date: 05/31/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/22/2023	263.53 *ton*	As-is		66.2	4,300.00	600.00	5,400.00		7.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.96	206.03	28.75	258.74	1,149.85

#4

Field name: #4

10/19/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 10/19/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	244.00 *ton*	As-is		71.6	5,200.00	1,100.00	7,400.00		13.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.20	126.88	26.84	180.56	907.78

05/31/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 05/31/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/22/2023	394.00 *ton*	As-is		66.4	4,500.00	500.00	5,900.00		7.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	19.70	177.30	19.70	232.46	992.88

NUTRIENT BUDGET

A. LAND APPLICATIONS

#1 - 10/18/2022: Oats, silage-soft dough

Field name: #1

Crop: Oats, silage-soft dough Plant date: 10/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/04/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	41.91	2.53	47.39	612.44	205,000.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	308.78	3,442,000.00 *gal*
Application event totals		41.91	2.53	47.39	921.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
12/30/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	32.87	1.99	37.17	480.36	160,788.00 *gal*
Application event totals		32.87	1.99	37.17	480.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	33.57	4.45	45.80	326.78	142,916.00 *gal*
Application event totals		33.57	4.45	45.80	326.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	32.33	4.29	44.10	314.69	137,626.00 *gal*
Application event totals		32.33	4.29	44.10	314.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	47.30	3.14	60.94	549.54	166,715.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	233.47	2,602,480.00 *gal*
Application event totals		47.30	3.14	60.94	783.00	

#1 - 05/31/2023: Corn, silage

Field name: #1

Crop: Corn, silage

Plant date: 05/31/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	128.14	17.72	53.16	2,835.46	160.00 *ton*
Application event totals		128.14	17.72	53.16	2,835.46	

05/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	303.45	3,382,640.00 *gal*
Application event totals		0.00	0.00	0.00	303.45	

06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	57.64	3.83	74.25	669.64	203,150.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	268.61	2,994,300.00 *gal*
Application event totals		57.64	3.83	74.25	938.25	

07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	31.62	8.89	110.45	409.73	355,787.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	238.99	2,664,040.00 *gal*
Application event totals		31.62	8.89	110.45	648.72	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	37.79	10.63	131.99	489.64	425,180.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	222.87	2,484,320.00 *gal*
Application event totals		37.79	10.63	131.99	712.51	

#1 - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	217.11	2,420,160.00 *gal*
Application event totals		0.00	0.00	0.00	217.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	20.01	5.63	69.90	259.32	225,180.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	210.90	2,350,960.00 *gal*
Application event totals		20.01	5.63	69.90	470.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	215.32	2,400,180.00 *gal*
Application event totals		0.00	0.00	0.00	215.32	

#2 - 10/18/2022: Oats, silage-soft dough

Field name: #2

Crop: Oats, silage-soft dough

Plant date: 10/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/04/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	29.59	1.79	33.46	432.39	115,787.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	304.84	2,718,480.00 *gal*
Application event totals		29.59	1.79	33.46	737.23	

#2 - 10/18/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	35.92	4.77	48.99	349.60	122,316.00 *gal*
Application event totals		35.92	4.77	48.99	349.60	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	46.46	6.16	63.38	452.25	158,231.00 *gal*
Application event totals		46.46	6.16	63.38	452.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	38.20	5.07	52.10	371.79	130,080.00 *gal*
Application event totals		38.20	5.07	52.10	371.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	34.25	2.28	44.12	397.85	96,558.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	233.28	2,080,352.00 *gal*
Application event totals		34.25	2.28	44.12	631.13	

#2 - 05/31/2023: Corn, silage

Field name: #2

Crop: Corn, silage

Plant date: 05/31/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	128.14	17.72	53.16	2,835.46	128.00 *ton*
Application event totals		128.14	17.72	53.16	2,835.46	

#2 - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	299.62	2,671,968.00 *gal*
Application event totals		0.00	0.00	0.00	299.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	78.60	5.22	101.25	913.13	221,615.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	275.12	2,453,424.00 *gal*
Application event totals		78.60	5.22	101.25	1,188.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	254.79	2,272,144.00 *gal*
Application event totals		0.00	0.00	0.00	254.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	31.95	8.98	111.60	414.00	287,595.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	233.63	2,083,488.00 *gal*
Application event totals		31.95	8.98	111.60	647.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	20.95	5.89	73.19	271.51	188,615.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	221.97	1,979,440.00 *gal*
Application event totals		20.95	5.89	73.19	493.48	

#2 - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	214.79	1,915,440.00 *gal*
Application event totals		0.00	0.00	0.00	214.79	

09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	22.48	6.32	78.51	291.26	202,330.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	217.58	1,940,304.00 *gal*
Application event totals		22.48	6.32	78.51	508.83	

#3 - 10/18/2022: Oats, silage-soft dough

Field name: #3

Crop: Oats, silage-soft dough Plant date: 10/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/04/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	39.08	2.36	44.18	570.98	105,118.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	299.44	1,835,845.00 *gal*
Application event totals		39.08	2.36	44.18	870.42	

01/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	37.22	4.94	50.78	362.33	87,155.00 *gal*
Application event totals		37.22	4.94	50.78	362.33	

#3 - 10/18/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	31.02	4.12	42.32	301.96	72,633.00 *gal*
Application event totals		31.02	4.12	42.32	301.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/28/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	42.35	5.62	57.77	412.20	99,150.00 *gal*
Application event totals		42.35	5.62	57.77	412.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	40.53	2.69	52.21	470.81	78,557.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	231.45	1,419,022.00 *gal*
Application event totals		40.53	2.69	52.21	702.26	

#3 - 05/31/2023: Corn, silage

Field name: #3

Crop: Corn, silage

Plant date: 05/31/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	128.14	17.72	53.16	2,835.46	88.00 *ton*
Application event totals		128.14	17.72	53.16	2,835.46	

#3 - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	64.77	4.31	83.44	752.45	125,550.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	306.84	1,881,220.00 *gal*
Application event totals		64.77	4.31	83.44	1,059.29	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	271.08	1,661,957.00 *gal*
Application event totals		0.00	0.00	0.00	271.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	43.16	12.14	150.77	559.30	267,115.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	248.88	1,525,865.00 *gal*
Application event totals		43.16	12.14	150.77	808.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	226.49	1,388,585.00 *gal*
Application event totals		0.00	0.00	0.00	226.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	28.62	8.05	99.95	370.79	177,088.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	221.07	1,355,376.00 *gal*
Application event totals		28.62	8.05	99.95	591.86	

#3 - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	212.46	1,302,565.00 *gal*
Application event totals		0.00	0.00	0.00	212.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	19.48	5.48	68.04	252.41	120,550.00 *gal*
T.I.D Canal	Surface water	0.00	0.00	0.00	217.25	1,331,935.00 *gal*
Application event totals		19.48	5.48	68.04	469.66	

#4 - 10/19/2022: Oats, silage-soft dough

Field name: #4

Crop: Oats, silage-soft dough

Plant date: 10/19/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	163.52	0.72	152.05	2,904.66	110.00 *ton*
Application event totals		163.52	0.72	152.05	2,904.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/07/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	303.62	3,384,540.00 *gal*
Application event totals		0.00	0.00	0.00	303.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	239.04	2,664,600.00 *gal*
Application event totals		0.00	0.00	0.00	239.04	

#4 - 05/31/2023: Corn, silage

Field name: #4

Crop: Corn, silage

Plant date: 05/31/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	240.26	33.23	99.68	5,316.48	300.00 *ton*
Application event totals		240.26	33.23	99.68	5,316.48	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	295.89	3,298,320.00 *gal*
Application event totals		0.00	0.00	0.00	295.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	282.26	3,146,440.00 *gal*
Application event totals		0.00	0.00	0.00	282.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	358.06	3,991,320.00 *gal*
Application event totals		0.00	0.00	0.00	358.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	344.49	3,840,140.00 *gal*
Application event totals		0.00	0.00	0.00	344.49	

#4 - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	266.82	2,974,300.00 *gal*
Application event totals		0.00	0.00	0.00	266.82	

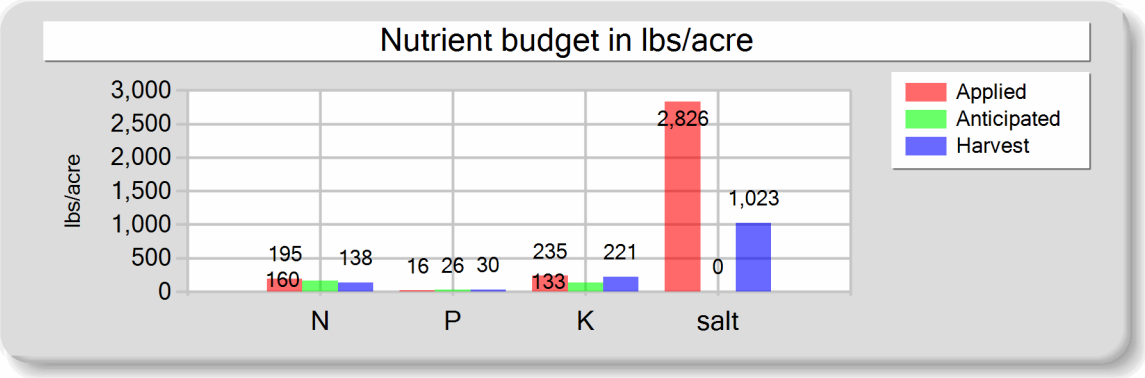
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
T.I.D Canal	Surface water	0.00	0.00	0.00	254.81	2,840,400.00 *gal*
Application event totals		0.00	0.00	0.00	254.81	

B. NUTRIENT BUDGET

#1 - 10/18/2022: Oats, silage-soft dough

Field name: #1 Crop: Oats, silage-soft dough Plant date: 10/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	187.99	16.41	235.39	2,283.80
Fresh water	0.00	0.00	0.00	542.24
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	194.99	16.41	235.39	2,826.04
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	138.31	30.18	221.30	1,023.02
Nutrient balance	56.68	-13.77	14.09	1,803.02
Applied to removed ratio	1.41	0.54	1.06	2.76

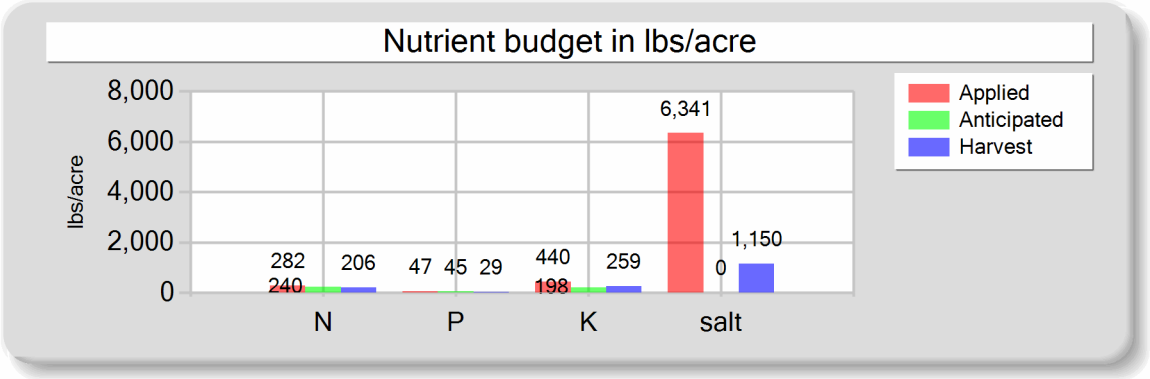
Fresh water applied	
6,044,480.00	*gallons*
222.60	*acre-inches*
11.13	*inches/acre*

Process wastewater applied	
813,045.00	*gallons*
29.94	*acre-inches*
1.50	*inches/acre*

Total harvests for the crop	
1	*harvests*

#1 - 05/31/2023: Corn, silage

Field name: #1 Crop: Corn, silage Plant date: 05/31/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	128.14	17.72	53.16	2,835.46
Process wastewater	147.06	28.98	386.60	1,828.33
Fresh water	0.00	0.00	0.00	1,677.25
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	282.20	46.70	439.76	6,341.03
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	206.03	28.75	258.74	1,149.84
Nutrient balance	76.17	17.95	181.02	5,191.19
Applied to removed ratio	1.37	1.62	1.70	5.51

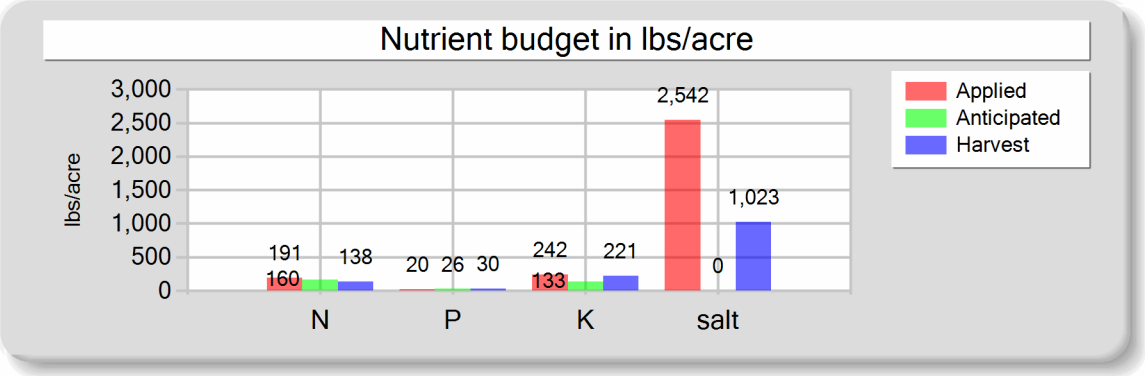
Fresh water applied	
18,696,600.00	*gallons*
688.53	*acre-inches*
34.43	*inches/acre*

Process wastewater applied	
1,209,297.00	*gallons*
44.53	*acre-inches*
2.23	*inches/acre*

Total harvests for the crop	
1	*harvests*

#2 - 10/18/2022: Oats, silage-soft dough

Field name: #2 Crop: Oats, silage-soft dough Plant date: 10/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	184.41	20.06	242.05	2,003.88
Fresh water	0.00	0.00	0.00	538.12
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	191.41	20.06	242.05	2,542.01
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	138.32	30.18	221.31	1,023.05
Nutrient balance	53.09	-10.12	20.74	1,518.95
Applied to removed ratio	1.38	0.66	1.09	2.48

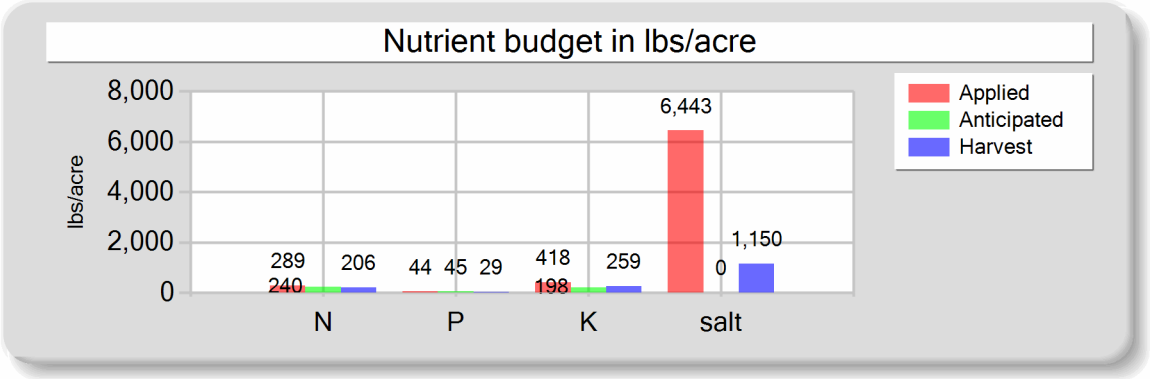
Fresh water applied	
4,798,832.00	*gallons*
176.72	*acre-inches*
11.05	*inches/acre*

Process wastewater applied	
622,972.00	*gallons*
22.94	*acre-inches*
1.43	*inches/acre*

Total harvests for the crop	
1	*harvests*

#2 - 05/31/2023: Corn, silage

Field name: #2 Crop: Corn, silage Plant date: 05/31/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	128.14	17.72	53.16	2,835.46
Process wastewater	153.98	26.42	364.56	1,889.90
Fresh water	0.00	0.00	0.00	1,717.50
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	289.12	44.14	417.72	6,442.85
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	206.03	28.75	258.73	1,149.83
Nutrient balance	83.09	15.40	158.99	5,293.02
Applied to removed ratio	1.40	1.54	1.61	5.60

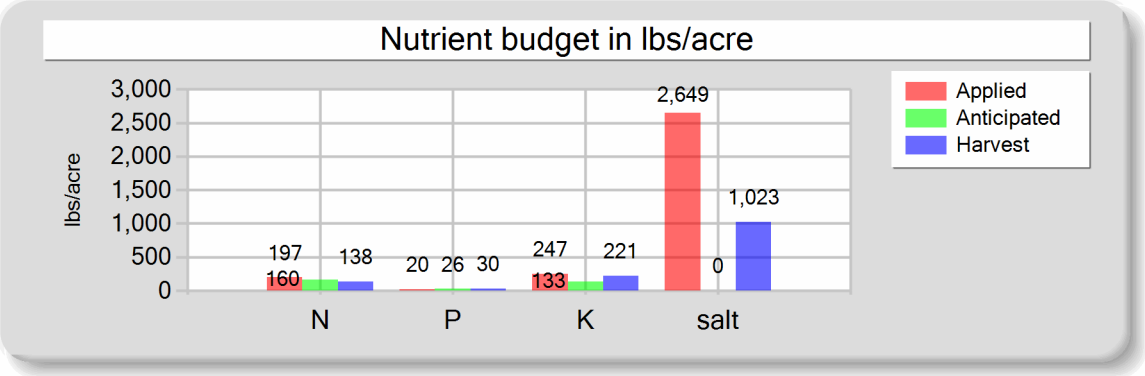
Fresh water applied	
15,316,208.00	*gallons*
564.04	*acre-inches*
35.25	*inches/acre*

Process wastewater applied	
900,155.00	*gallons*
33.15	*acre-inches*
2.07	*inches/acre*

Total harvests for the crop	
1	*harvests*

#3 - 10/18/2022: Oats, silage-soft dough

Field name: #3 Crop: Oats, silage-soft dough Plant date: 10/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	190.20	19.73	247.25	2,118.28
Fresh water	0.00	0.00	0.00	530.89
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	197.20	19.73	247.25	2,649.17
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	138.31	30.18	221.30	1,022.99
Nutrient balance	58.89	-10.45	25.96	1,626.18
Applied to removed ratio	1.43	0.65	1.12	2.59

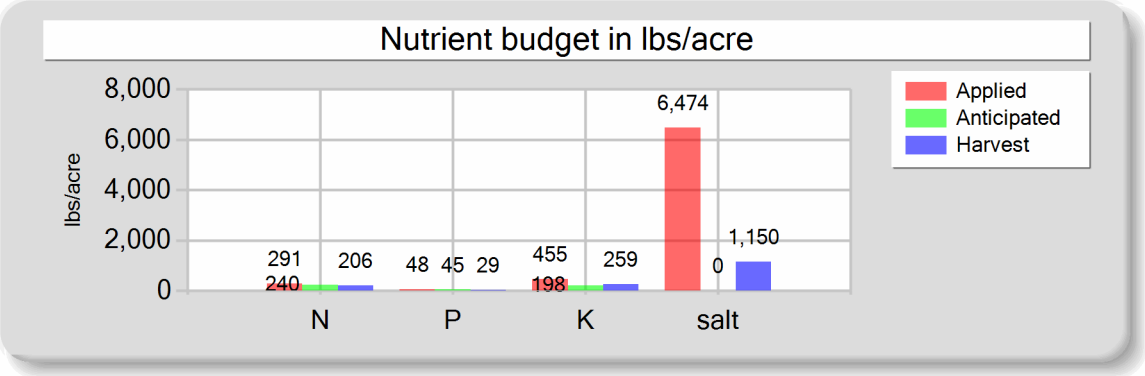
Fresh water applied	
3,254,867.00	*gallons*
119.87	*acre-inches*
10.90	*inches/acre*

Process wastewater applied	
442,613.00	*gallons*
16.30	*acre-inches*
1.48	*inches/acre*

Total harvests for the crop	
1	*harvests*

#3 - 05/31/2023: Corn, silage

Field name: #3 Crop: Corn, silage Plant date: 05/31/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	128.14	17.72	53.16	2,835.46
Process wastewater	156.03	29.97	402.20	1,934.95
Fresh water	0.00	0.00	0.00	1,704.06
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	291.17	47.69	455.36	6,474.47
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	206.03	28.75	258.74	1,149.85
Nutrient balance	85.13	18.94	196.62	5,324.61
Applied to removed ratio	1.41	1.66	1.76	5.63

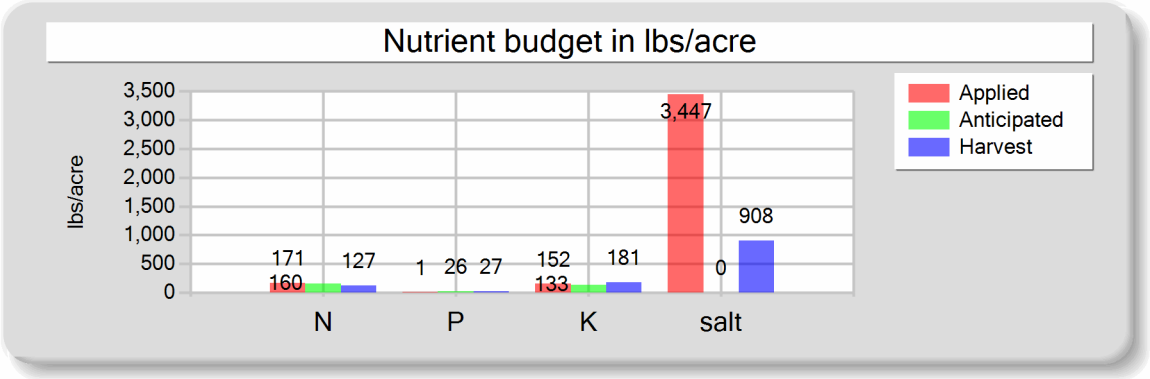
Fresh water applied	
10,447,503.00	*gallons*
384.75	*acre-inches*
34.98	*inches/acre*

Process wastewater applied	
690,303.00	*gallons*
25.42	*acre-inches*
2.31	*inches/acre*

Total harvests for the crop	
1	*harvests*

#4 - 10/19/2022: Oats, silage-soft dough

Field name: #4 Crop: Oats, silage-soft dough Plant date: 10/19/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	163.52	0.72	152.05	2,904.66
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	542.66
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	170.52	0.72	152.05	3,447.32
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	126.88	26.84	180.56	907.78
Nutrient balance	43.64	-26.12	-28.51	2,539.54
Applied to removed ratio	1.34	0.03	0.84	3.80

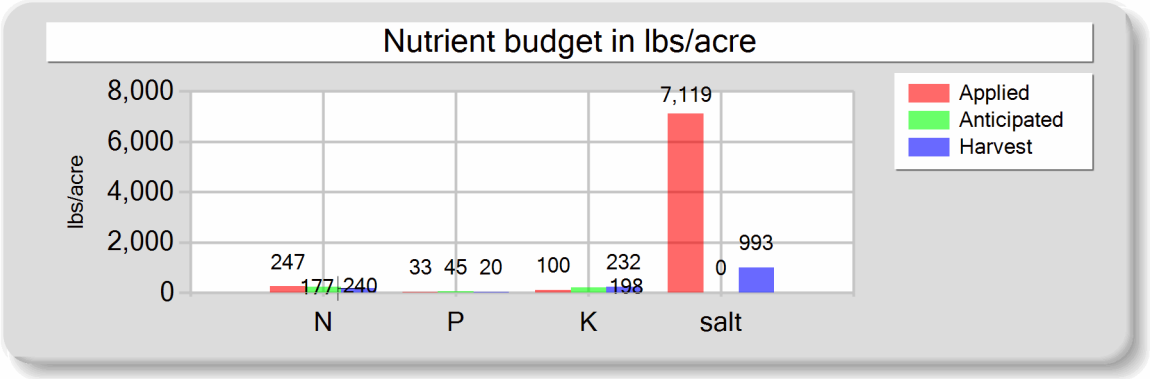
Fresh water applied	
6,049,140.00	*gallons*
222.77	*acre-inches*
11.14	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

#4 - 05/31/2023: Corn, silage

Field name: #4 Crop: Corn, silage Plant date: 05/31/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	240.26	33.23	99.68	5,316.48
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	1,802.33
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	247.26	33.23	99.68	7,118.81
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	177.30	19.70	232.46	992.88
Nutrient balance	69.96	13.53	-132.78	6,125.93
Applied to removed ratio	1.39	1.69	0.43	7.17

Fresh water applied	
20,090,920.00	*gallons*
739.88	*acre-inches*
36.99	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

2022 Fall Corral Solids

Sample and source description: 2022 Fall Corral Solids
Sample date: 11/15/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 34.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,800.00	100.00	21,200.00							40.50
DL	100.00	100.00	100.00							0.01

Spring Corral Solids

Sample and source description: Spring Corral Solids
Sample date: 05/03/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 57.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,800.00	2,600.00	7,800.00							41.60
DL	100.00	100.00	100.00							0.01

Fall Corral Solids

Sample and source description: Fall Corral Solids
Sample date: 10/23/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 19.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,000.00	3,700.00	30,100.00	9,700.00	5,200.00	6,800.00	3,800.00	952.00		24.30
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00		0.01

B. PROCESS WASTEWATER ANALYSES

2022 WWSQ4

Sample and source description: 2022 WWSQ4

Sample date: 12/13/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	490.00	339.00	0.00	0.00	29.60	554.00								6,520.00	7,160
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ1

Sample and source description: WWSQ1

Sample date: 03/22/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	563.00	398.00	0.00	0.00	74.70	768.00								7,250.00	5,480
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ2

Sample and source description: WWSQ2

Sample date: 06/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	680.00	406.00	0.00	0.00	45.20	876.00								8,170.00	7,900
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWSQ3

Sample and source description: WWSQ3

Sample date: 09/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	213.00	92.40	0.00	0.00	59.90	744.00	81.40	74.00	164.00	1,760.00	0.00	111.00	243.00	4,190.00	2,760
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	1.00	10

WWSQ4

Sample and source description: WWSQ4

Sample date: 12/12/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	624.00	459.00	0.00	0.00	29.00	730.00								8,410.00	7,520
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

T.I.D Canal

Canal

Sample description: Canal

Sample date: 08/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								0.00	215
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

#1 - 10/18/2022: Oats, silage-soft dough

#1 - 10/18/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,500.00	1,200.00	8,800.00		11.30
DL	100.00	100.00	100.00		0.01

#1 - 05/31/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,300.00	600.00	5,400.00		7.10
DL	100.00	100.00	100.00		0.01

#2 - 10/18/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,500.00	1,200.00	8,800.00		11.30
DL	100.00	100.00	100.00		0.01

#2 - 05/31/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,300.00	600.00	5,400.00		7.10
DL	100.00	100.00	100.00		0.01

#3 - 10/18/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,500.00	1,200.00	8,800.00		11.30
DL	100.00	100.00	100.00		0.01

#3 - 05/31/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,300.00	600.00	5,400.00		7.10
DL	100.00	100.00	100.00		0.01

#4 - 10/19/2022: Oats, silage-soft dough

Winter Forage

Sample and source description: Winter Forage

Sample date: 06/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,200.00	1,100.00	7,400.00		13.10
DL	100.00	100.00	100.00		0.01

#4 - 05/31/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/25/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,500.00	500.00	5,900.00		7.50
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

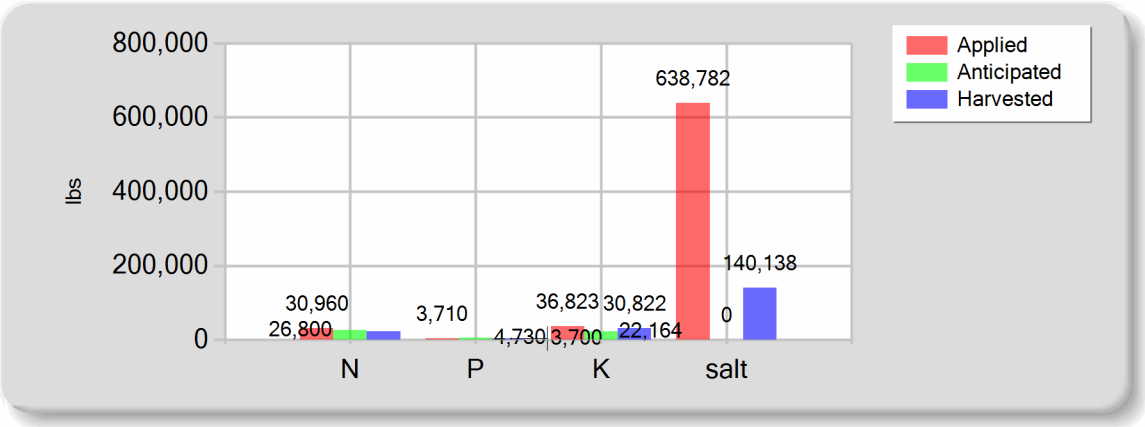
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

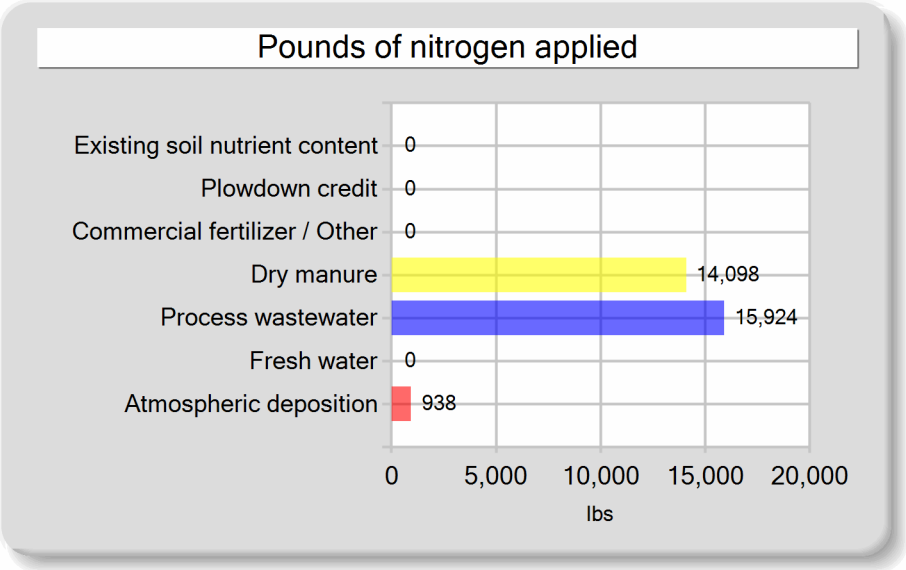
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

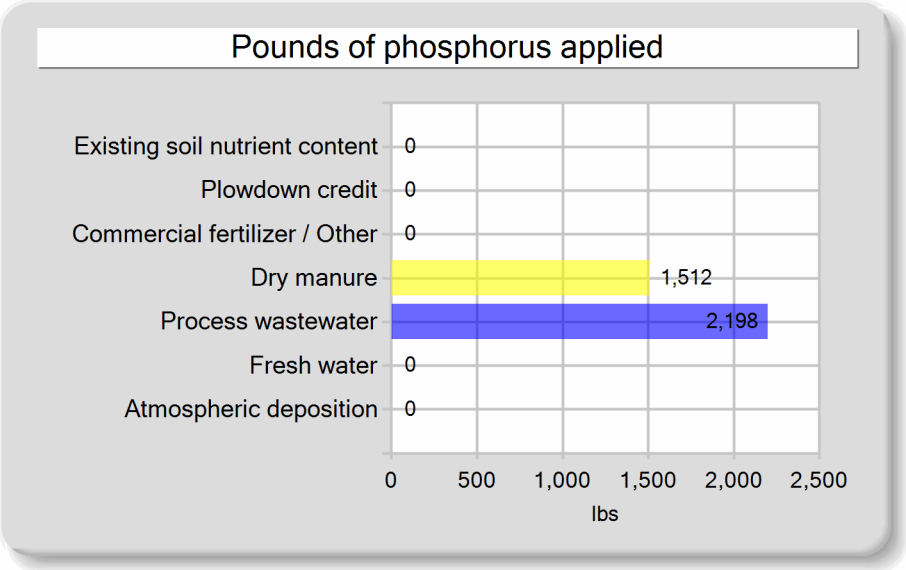
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	14,098.33	1,511.82	7,533.35	297,689.23
Process wastewater	15,923.75	2,198.12	29,289.39	189,128.64
Fresh water	0.00	0.00	0.00	151,964.02
Atmospheric deposition	938.00	0.00	0.00	0.00
Total nutrients applied	30,960.08	3,709.94	36,822.74	638,781.90
Anticipated crop nutrient removal	26,800.00	4,730.20	22,163.60	0.00
Actual crop nutrient removal	22,267.81	3,700.33	30,822.23	140,137.82
Nutrient balance	8,692.28	9.61	6,000.51	498,644.08
Applied to removed ratio	1.39	1.00	1.19	4.56

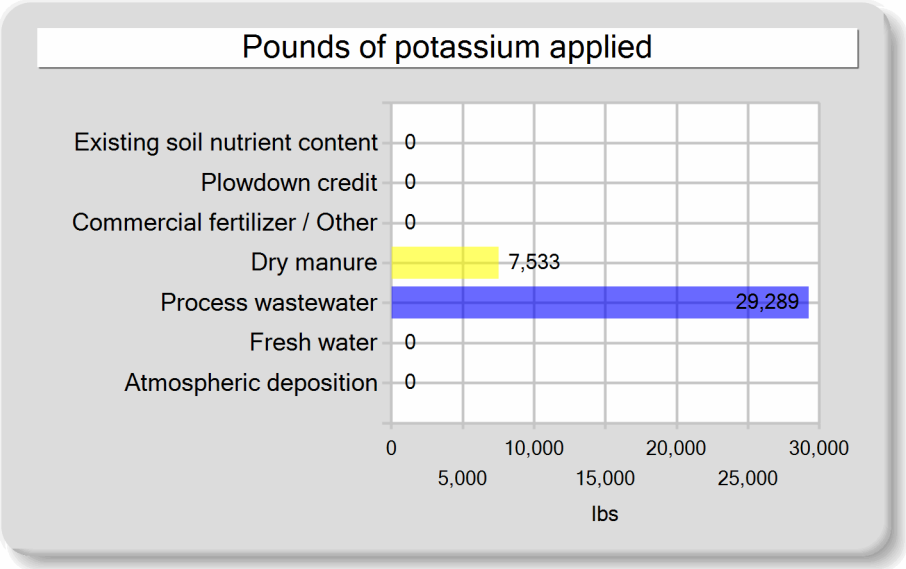
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

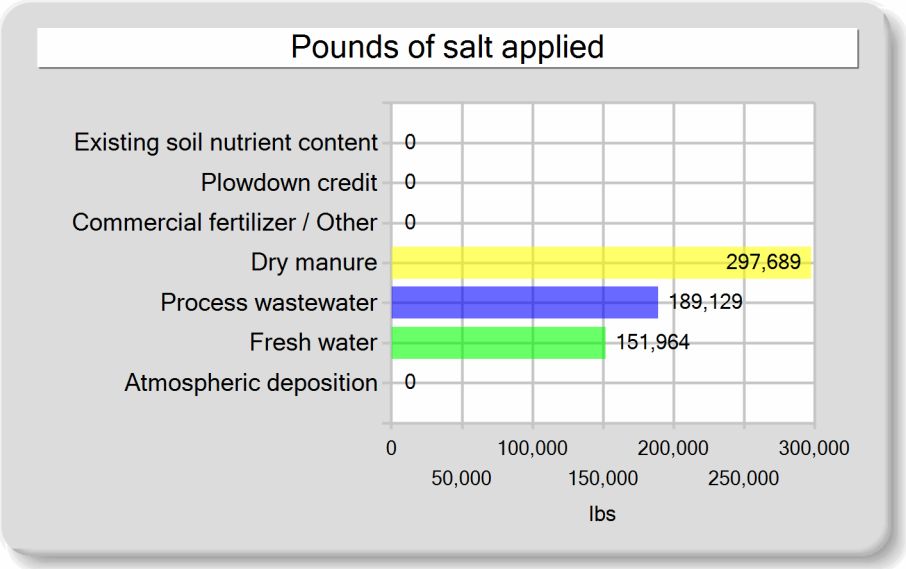


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

David King (Jun 21, 2024 06:09 PDT)	David King (Jun 21, 2024 06:09 PDT)
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
David King	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 21, 2024	Jun 21, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
King Dairy
5524 S. Central Ave.
Turlock, CA 95380

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/30/23 2:35 PM
SUBMITTED BY: KJ
ANALYZED DATE: 10/31/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 21.3c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW #1	T3303043A	Electrical Conductivity (EC)	EPA 120.1 and	1.12	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	46.7	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	717	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	96.0	95.8 (T3303043A)	94.6 (T3303043A)	1.26
Electrical Conductivity (EC)	97.1			0.309
Nitrate Nitrogen (NO3-N)	97.5	98.5 (T3303041A)	101 (T3303041A)	2.29

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

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40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
King Dairy
5524 S. Central Ave.
Turlock, CA 95380

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 10/30/23 2:35 PM
SUBMITTED BY: KJ
ANALYZED DATE: 10/31/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 21.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW #2	T3303043B	Electrical Conductivity (EC)	EPA 120.1 and	1.11	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	47.1	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	712	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	96.0	95.8 (T3303043A)	94.6 (T3303043A)	1.26
Electrical Conductivity (EC)	97.1			0.309
Nitrate Nitrogen (NO3-N)	97.5	98.5 (T3303041A)	101 (T3303041A)	2.29

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: King Dairy
Mailing address:
(not facility address)
Phone#:

Authorized Copy Released To: F & R Ag Services

Sampled By: K.Jones
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
							Analysis				
1	10/30	10:50	Domestic Wells	DW #1	21.3		X				
2	↓	11:00	Domestic Wells	DW #2	21.0		X				
3											
4											
5											
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged In By:	LABORATORY NUMBER
1st		F & R Ag Services				
2nd		DENELE ANALYTICAL, INC.	10/30/23 14:35			T3303043

Required to be submitted to any agency? (YES / NO)

2023 Annual Report- King Dairy

Final Audit Report 2024-06-21

Created:	2024-06-19
By:	Kristen Jones (kjones@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAPiVxKAmdeatJ8buBABsVb6x65nembQgc

"2023 Annual Report- King Dairy" History

Document created by Kristen Jones (kjones@fragservices.com)
2024-06-19 - 10:33:26 PM GMT

Document emailed to patti_cakes2@yahoo.com for signature
2024-06-19 - 10:34:00 PM GMT

Email viewed by patti_cakes2@yahoo.com
2024-06-19 - 10:44:52 PM GMT

Signer patti_cakes2@yahoo.com entered name at signing as David King
2024-06-21 - 1:09:13 PM GMT

Document e-signed by David King (patti_cakes2@yahoo.com)
Signature Date: 2024-06-21 - 1:09:15 PM GMT - Time Source: server

Agreement completed.
2024-06-21 - 1:09:15 PM GMT